

NORTH SEA STATION 63104, United Kingdom

WMO: 995270

Lat: 61.200N

Lon: 1.600E

Elev: 30

StdP: 100.97

Time Zone: 0.00 (EUW)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.1	1.0	-7.6	2.0	2.5	-6.2	2.2	3.1	22.5	4.5	20.8	4.9	9.1	340	0.908

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	18.6	15.0	17.2	14.5	16.2	13.9	16.0	17.7	15.2	16.7	14.5	15.8	5.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	10.9	16.7	14.5	10.3	15.9	13.7	9.8	15.2	45.0	17.9	42.7	16.9	40.6	15.8	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	18.1	15.9	-1.2	21.9	1.0	2.2	-1.9	23.5	-2.5	24.8	-3.0	26.1	-3.8	27.7
			-2.5	17.9	0.9	1.3	-3.1	18.8	-3.7	19.6	-4.2	20.3	-4.9	21.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	5.6	5.0	5.4	6.8	9.0	11.4	13.4	14.3	12.6	9.7	8.0	6.2
	DBStd	3.67	1.72	2.07	2.01	1.84	1.76	1.81	1.82	1.63	1.63	1.84	1.81	2.22
	HDD10.0	773	138	140	142	98	41	6	0	0	2	28	61	118
	HDD18.3	3422	396	373	401	347	289	209	153	127	173	269	309	376
	CDD10.0	393	0	0	0	1	10	47	106	132	78	17	2	0
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.3	10.4	8.9	8.8	7.9	6.9	6.5	6.6	6.6	7.8	8.9	9.6	10.1
Precipitation	PrecAvg	1341	161	132	127	72	48	65	81	100	117	141	139	156
	PrecMax	1652	252	257	237	123	83	132	153	162	190	216	220	240
	PrecMin	954	76	58	46	26	22	26	40	66	48	58	43	65
	PrecStd	165	48	53	46	23	16	24	22	25	34	41	39	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	9.7	9.9	13.3	15.5	18.9	20.9	21.0	17.9	14.2	11.8	10.6
		MCWB	7.9	8.3	8.0	9.2	11.3	13.6	16.3	15.9	14.5	11.8	10.3	9.2
	2%	DB	9.1	8.9	9.1	10.8	13.6	16.4	18.7	18.5	16.0	13.1	11.1	10.1
		MCWB	7.6	7.5	7.4	7.9	10.2	12.8	15.4	15.2	13.9	11.0	9.7	8.9
	5%	DB	8.6	8.4	8.5	9.7	12.3	14.9	17.0	17.3	15.0	12.4	10.7	9.7
		MCWB	7.1	7.0	6.9	7.5	9.4	12.0	14.5	14.7	13.5	10.4	9.3	8.4
	10%	DB	8.1	7.8	7.9	8.9	11.2	13.6	15.9	16.3	14.4	11.8	10.2	9.2
		MCWB	6.5	6.5	6.4	7.2	9.0	11.2	13.9	14.2	13.1	9.9	8.9	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	9.0	8.7	9.8	12.3	15.2	17.2	17.0	16.1	12.9	11.1	10.0
		MCDB	9.0	9.3	9.3	12.5	14.8	17.6	19.6	19.0	16.5	13.6	11.3	10.1
	2%	WB	8.1	8.2	8.0	8.7	10.8	13.3	15.9	16.1	14.8	11.7	10.4	9.4
		MCDB	8.7	8.6	8.7	9.8	12.4	15.7	17.6	17.9	15.4	12.4	10.8	9.8
	5%	WB	7.5	7.4	7.4	8.1	10.1	12.4	15.1	15.4	14.1	10.8	9.9	8.7
		MCDB	8.3	8.1	8.2	9.2	11.5	14.4	16.9	17.0	14.6	11.7	10.4	9.5
	10%	WB	6.8	6.6	6.7	7.6	9.5	11.7	14.2	14.6	13.6	10.3	9.4	8.1
		MCDB	7.8	7.6	7.7	8.7	10.8	13.2	15.7	15.9	14.3	11.4	10.0	9.0
Mean Daily Temperature Range	5% DB	MDBR	2.7	2.8	3.0	3.1	3.5	3.3	3.1	3.3	2.7	2.6	2.5	2.8
		MCDBR	3.1	3.3	3.3	4.3	5.0	5.2	4.7	4.9	3.6	3.0	2.6	2.7
		MCWBR	3.5	3.2	3.2	3.2	3.4	3.3	2.8	3.1	2.7	2.8	2.8	3.2
	5% WB	MCDBR	3.0	2.7	3.0	3.8	4.3	4.5	4.3	4.2	2.9	2.6	2.6	2.7
		MCWBR	3.7	3.3	3.4	3.2	3.2	3.1	2.7	3.1	2.5	2.9	3.1	3.6
Clear-Sky Solar Irradiance	taub	0.334	0.355	0.368	0.380	0.369	0.361	0.380	0.369	0.364	0.345	0.319	0.380	
	taud	2.062	2.185	2.267	2.306	2.409	2.472	2.437	2.466	2.431	2.308	2.128	2.290	
	Ebn at Noon	342	593	738	809	853	868	838	818	747	607	359	213	
	Edh at Noon	44	72	91	106	103	98	100	90	78	62	40	28	
All-Sky Solar Radiation	RadAvg	0.19	0.67	1.80	3.50	4.89	5.01	4.54	3.63	2.20	1.03	0.30	0.12	
	RadStd	0.02	0.08	0.20	0.28	0.35	0.55	0.41	0.34	0.18	0.13	0.04	0.01	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (22 neighbors)		N/A	+0.65	+0.62	N/A	N/A	N/A	-102	-134	N/A	N/A	

Nomenclature: See separate page